

FAISS Vector Search Fundamentals

FAISS, short for Facebook AI Similarity Search, is a library developed by Meta AI Research for efficient similarity search and clustering of dense vectors. It is widely used in production machine learning systems because it can handle datasets containing millions or even billions of vectors while still returning nearest-neighbor results in milliseconds. FAISS supports several index types, ranging from simple flat indexes that perform an exact brute-force search, to more advanced approximate indexes such as IVF (Inverted File) and HNSW (Hierarchical Navigable Small World graphs), which trade a small amount of accuracy for significantly faster search speed on very large collections.

Embedding Models for Text

Before FAISS can search over text data, that text must first be converted into numerical vectors using an embedding model. Sentence-Transformers is a popular open-source Python library that provides many pretrained models for this purpose, including all-MiniLM-L6-v2, a compact model that produces 384-dimensional embeddings while remaining fast enough to run on a laptop CPU. These embeddings are designed so that sentences with similar meaning are placed close together in vector space, which is what allows semantic search to find conceptually related passages even when they do not share the same exact words as the query.

Document Chunking Best Practices

When working with long PDF reports, splitting the document into smaller overlapping chunks before embedding is essential for good retrieval quality. If chunks are too large, the resulting embedding becomes a blurry average of many different ideas, making it harder to match a specific query. If chunks are too small, important context can be lost. A chunk size of roughly 1000 characters with a 200 character overlap is a commonly recommended starting point, since it keeps each chunk focused on a single topic while ensuring that sentences near chunk boundaries still appear in at least one complete chunk.

